Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Fiscal year 3.1 · Direct calculations

In an orthonormous basis of the plan, vectors are considered $\vec{u} = (1; -2)$ and $\vec{v} = (4; 3)$.

1. Calculate the scalar product $\vec{u} \cdot \vec{v}$, then the norms $\|\vec{u}\|$ and $\|\vec{v}\|$.

2. Calculate the coordinates of $\vec{u} + \vec{v}$ and deduct from it $\|\vec{u} + \vec{v}\|^2$.

3. To find this last result with the help of identity involving $\vec{u} \cdot \vec{v}$.

Exercise 3.2 · A right triangle?

The plan is equipped with an orthonormal coordinate system. $A(1; 1)$, $B(4; 2)$ and $C(0; 4)$.

1. Calculate the coordinates of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then their scalar product.

2. Deducing that the triangle ABC is rectangle; specify at which top.

3. Calculate the lengths of the two perpendicular sides and then the triangle's area in diode units.

Exercise 3.3 · Length without reference

The Quadrilateral $ABCD$ is a parallelogram such as $AB = 4$, $AD = 3$ and $\widehat{BAD} = 60°$. The lengths are expressed in the same unit.

1. Make a figure and calculate the scalar product $\overrightarrow{AB} \cdot \overrightarrow{AD}$.

2. Determine exact diagonal lengths $[AC]$ and $[BD]$. Explain how vectors corresponding to the diagonals expressed from $\overrightarrow{AB}$ and $\overrightarrow{AD}$.

3. Indicate which diagonal is the longest and check that this result is consistent with your figure.

Exercise 3.4 · Find an angle

Two non-zero vectors $\vec{u}$ and $\vec{v}$ check $\|\vec{u}\| = 2$, $\|\vec{v}\| = 6$ and $\vec{u} \cdot \vec{v} = -6$. Note θ their angle no oriented, within 0 ° and 180°.

1. Determine the exact value of cos θ, then the measurement of θ.

2. Explain how the sign of the scalar product allowed predicting that the angle would be obtuse.

Exercise 3.5 · A variable location

Points A and B are distinct and fixed. I the midpoint of $[AB]$. For a real k set, we're looking All points M of the plan that verifies $\overrightarrow{MA} \cdot \overrightarrow{MB} = k$.

1. Express this scalar product according to MI and AB using Chasles' relationship and A scalar identity.
2. Describe precisely the desired set according to the value of k. For each case, indicate whether it is empty, reduced to a point or constituted by a circle; then give its characteristics.
3. Explain the role of whether a squared distance is positive or zero.

Exercise 3.6 · Force Work

An object moves in a straight line of A To B over a distance of 4 m. It is subjected to a force constant $\vec{F}$ 50 N. The work of this force is defined by $W = \vec{F} \cdot AB$ and expressed in Joules.

1. Calculate W when the angle is between $\vec{F}$ and $\overrightarrow{AB}$ value 60°.
2. Resume calculation for an angle of 120 °.
3. Compare the signs of the two works and explain, in each situation, whether the force favours the Displacement or opposing.

Fiscal year 3.7 · Contrapositive

The following statement, valid for two vectors of the plan, is to be demonstrated:

$$\|\vec{u}+\vec{v}\|^2 \neq \|\vec{u}\|^2 + \|\vec{v}\|^2 \quad \Longrightarrow \quad \vec{u} \text{ and } \vec{v} \text{ are not orthogonal} \quad .$$

1. State the contraposition of this involvement.
2. Demonstrate this contraposition by means of a scalar identity, and then conclude on the initial statement.

Fiscal year 3.8 · Deferred review

A triangle ABC checks $AB = 3$, $AC = 4$ and $BC = 5$. We are looking $\overrightarrow{AB} \cdot \overrightarrow{AC}$ without using any coordinates.

1. Calculate this scalar product by a formula involving the three lengths.
2. Find the result by determining the nature of the triangle. Quote the theorem used and its numerical hypothesis.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 250, questions a and b. Calculate scalar product in orthonormized base and detail both coordinates products.

CQFD, Exercise 11 at 250, Questions 1, 2 and 3. Use successively the coordinates, Chasles' relationship and the fact that I is the midpoint.